

ASSIGNMENT



Inspiring Excellence

Submitted to : Sowik Ghosh (SVKG) Sir

Submitted by : Sheikh Md. Tahmid Hossain

ID : 222 99522

Section : 26

Assignment No: 1

Date : 29 / 10 / 25

Course Code : CSE 340

Answer to the Question No. 1

Performance via prediction means anticipating future data / values in order to avoid delays. In literal computer architecture terms refers to a strategy where it separately executes instructions to avoid waiting thus helps in increasing overall performance.

Example:-

```
if (x > 5) {          (1)
    statement 1      (2)
}
else {                (3)
    statement 2      (4)
}
```

When the processor encounters a line (1) it predicts which path / condition will likely be executed, and starts implementing the work and connects itself only if the prediction is wrong

Answer to the Question No. 3.2

Dependability means ensuring the system keeps working even if some parts of the system fail. This possible with the help of inclusion of redundant components which helps us to figure and detect the ones with failure. In case it also replaces them.

Example:

RAID C (Redundant Array of Independent Disk) is a configuration where data is duplicated across several hard drives. and thus boost a system's reliability.

If a drive fails, the system simply uses the data on the other drives to keep running. In a RAID 1 system data is mirrored across two hard drives and if one fails the

the other still holds an exact copy, keeping the system operational.

Answer to the Question No. 3 (a)

Amdahl's law is a formula which helps to predict the maximum possible speedup we can get by improving only a single part of a system.

$$T_{\text{improved}} = \frac{T_{\text{affected}}}{\text{Improve ment Factor}} + T_{\text{unaffected}}$$

(b)

We are required to relate Amdahl's law with the design patterns mentioned.

"(i) Performance the Common Case Fast" tends to enhance the performance

better than 'optimizing the rare case. which is only possible with measurement and caution. Amdahl's Law also supports the principle mathematically

Amdahl's Law states that,

$$\text{Speed Up} = \frac{1}{(1-a) + \frac{a}{b}}$$

a $\rightarrow$ Affected by improvement.

b $\rightarrow$ Speedup of that part

Example:

Let's say 80% time spent on a task

(Common case) . 20% on other operation

(Rare case)

$$\text{Speed Up} = \frac{1}{(1-0.8) + \frac{(0.8)}{2}} = 1.67$$

But if we double rare case instead

$$\text{Speed Up} = \frac{1}{(1-0.2) + \frac{0.2}{2}} = 1.11$$

So, improving common case $\rightarrow$ more benefit

Answer to the Question No. 4

We are required to narrate a scenario where increasing the throughput could also improve the response time.

Let's say a university's web portal gets extremely busy and saturated at 9:00 am when course registration starts. Each system request requires 50 ms of CPU time. Because of the saturation, the requests start to queue while waiting for CPU.

The IT team handles the saturation through adding another identical server behind a load balancer thus increasing system throughput which results in more requests finished per second. As there is no saturation now, the queue then shrinks and response time also drops.

Answer to the Question No.5

We know,

$$\text{Spec Ratio} = \frac{\text{Reference Time}}{\text{CPU Time}}$$

$$\text{Geometric Mean} = \sqrt[n]{\text{Spec Ratio 1} \times \text{Spec Ratio 2} \times \text{Spec Ratio 3} \dots}$$

Let say,

~~GNU C Compiler~~ → Reference Time = 1774

Perl interpreter

Execution Time = 627

$$\text{Spec Ratio} = \frac{1774}{627} = 2.83$$

Route planning spec Ratio → 3.89

Geometric Mean, of Perl Interpreter & Route planning,

$$= \sqrt{3.89 \times 2.83} \rightarrow 3.317$$

So, geometric mean is 3.317. If we used arithmetic mean

$$= \frac{(3.89 + 2.83)}{2} = 3.36$$

So, $(3.36 - 3.317) = 0.043$

So, if we used arithmetic it would imply that CPU is 4.3% faster.

Thus Handout specifies and represents that geometric mean should be used in case of comparing spec ratios

Answer to the Question No.6

we know,

$$\text{CPU Time} = \frac{\text{CPI}}{\text{Clock Rate}} \times \text{Instruction Count}$$

So, the respective factors are:-

Instruction Count $\rightarrow$ High level language translates into Machine Instruction thus it gets affected by programming language

CPI $\rightarrow$ language can increase or decrease CPI so yes it is affected.

No clock rate as it is a hardware / implementation property programming language ~~data~~ has no effect on it.